

09

Table 1 shows the volume of blood in a woman's left ventricle at different times during **one** second.

Table 1

Time / seconds	Volume of blood in left ventricle / cm ³
0.0	112
0.1	120
0.2	95
0.3	65
0.4	50
0.5	55
0.6	82
0.7	90
0.8	100
0.9	112
1.0	120

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Use **Table 1** to calculate the heart rate in beats per minute.

Tick (✓) **one** box next to the correct answer.

[1 mark]

60

☐

66.7

☐

75

☐

85.7

☐


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The stroke volume is the volume of blood pumped out of the left ventricle during one cardiac cycle.

Use **Table 1** to determine the stroke volume.

[1 mark]

Stroke volume = _____ cm³

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Some people produce a much higher ventricular blood pressure than normal. This can cause tissue fluid to build up outside the blood capillaries of these people.

Explain why.

[2 marks]

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Some drugs used to reduce high ventricular blood pressure cause widening of blood vessels.

Suggest how widening of blood vessels can reduce ventricular blood pressure.

[2 marks]

